Property	Type	Mandatory	Description
LBN-GTT-Partition-Key-Hint	String	No	A hint to determine which message queue the message will be sent to. Messages with the same hint will be sent to the same message queue and they will be processed in chronological order.

Sample Code

Request header example:

```
LBN-GTT-Message-Number: 9900000007778950LBN-GTT-Source-Id: QS8CLNT910LBN-GTT-Partition-Key-Hint:
```

Request Body

Here are sample request body of different scenarios:

- Create or update a tracked process without attachments
- Create a tracked process with attachments
- Add an additional attachment to an existing tracked process with attachments
- Update a tracked process to remove attachments

Response

Response Codes

The following HTTP response codes may occur:

Code	HTTP Description	Track and Trace Semantic
202	Accepted	Request has been received and will be processed later
400	Bad Request	The content of the request payload does not match the definition. The request payload should be qualified json string and contain altKey.
401	Unauthorized	There is no authentication passed with the request
403	Forbidden	The user does not access the right url to create the tracked process instance
404	Not found	The access url is inaccessible
405	Method not allowed	The method received in the request-line is known by the origin server but not supported by the target resource.
500	Internal Server Error	An unspecified error occurred during request processing
503	Service Unavailable	The request cannot be executed due to unavailability of required components, e.g. tenant onboarding hasn't been completed
504	Timeout	The server, while acting as a gateway or proxy, did not receive a timely response from an upstream server it needed to access in order to complete the request.

Response Body

The response body is empty.

Create or update a tracked process without attachments

Here is the sample request body of creating or updating a tracked process without attachments:

Sample Code

```
{
    "door": "",
    "initialPlannedDate": "2020-08-06",
    "destinationAddress": "Testname test$test street 123456$68766 HOCKENHEIM$GERMANY",
    "references": [{
        "referenceType_code": "TRACKING",
        "action_code": "ADD",
        "altKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY:0080045552",
        "validFrom": "2020-08-03T05:50:10Z",
        "validTo": "9999-12-30T23:00:00Z"
    }],
    "itemNo": "000020",
    "billofLading": "",
    "orderQuantity": 1.000,
    "executionStatus_code": "NOT_STARTED",
    "shippingPoint": "0001",
    "deliveryAltKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY:0080045552"
    "podStatus_code": "X",
    "weightUoM": "KG",
    "goodsIssueStatus_code": "A",
    "warehouseDescription": "",
    "quantityUoM": "EA",
    "pickingStatus_code": "C",
    "actualBusinessTimeZone": "CET",
    "upc": "5007509",
    "priority": 0,
    "volume": 1.000,
    "netWeight": 4.000,
    "incotermsVersion": "0001",
    "salesOrderItem_id": "42fdd0f4-a566-5b66-8e0c-d3d27dbb0272",
    "destinationTelephone": "13912341234",
    "materialNo": "HST_F1",
    "departureAddress": "Company$Dietmar-Hopp-Allee 16$69190 WALLDORF$GERMANY",
    "delivery_id": "56dcfddc-555c-5209-b7e8-fc863aaa5cf2",
    "destination": "0000010105",
    "destinationEmail": "test@163.com",
    "altKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY_IT:0080045552000020
    "modelNamespace": "com.lbngttapps.gtt.app.gttft1",
    "volumeUoM": "FT3",
    "actualTechnicalTimestamp": "2020-08-03T05:50:16.538Z",
    "dangerousGoods": "false",
    "deliveryNo": "0080045552",
    "plannedEventGroups": [
        {
            "group": "001",
```

```
                "plannedEventUpdateStrategy_code": "KEEP_REPORTED_WITH_DELETION"
            },
            {
                "group": "002",
                "plannedEventUpdateStrategy_code": "KEEP_REPORTED_WITHOUT_DELETION"
            }
        ],
        "plannedEvents": [
            {
                "plannedBusinessTimestamp": "2020-08-04T22:00:00Z",
                "plannedBusinessTimeZone": "CET",
                "plannedTechnicalTimestamp": "2020-08-04T22:00:00Z",
                "eventType": "com.lbngttapps.gtt.app.gttft1.Delivery.GoodsIssued",
                "locationAltKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:Location:ShippingPoin
                "group": "001"
            },
            {
                "plannedBusinessTimestamp": "2020-08-05T22:00:00Z",
                "plannedBusinessTimeZone": "CET",
                "plannedTechnicalTimestamp": "2020-08-04T22:00:00Z",
                "eventType": "com.lbngttapps.gtt.app.gttft1.Delivery.GoodsReceipt",
                "locationAltKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:Location:Customer:000
                "group": "002"
            }
        ],
        "incoterms_code": "CIF",
        "materialDescription": "Laptop",
        "warehouseNo": "",
        "actualBusinessTimestamp": "2020-08-03T05:50:10Z",
        "grossWeight": 5.000,
        "salesOrderItemAltKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:SALES_ORDER_ITEM:000007
        "incotermsLocation1": "",
        "packingStatus_code": "X",
        "revisedPlannedDate": "2020-08-06",
        "messageSourceType": "TM"
    }
```


Create a tracked process with attachments

You can create a tracked process with attachments in two formats: form-data format or JSON format.

Form-data Format

Request Header

`Content-Type: multipart/form-data`

Request Body

Here is the sample request body:

Sample Code

event:


```
{
    "door": "",
    "initialPlannedDate": "2020-08-06",
    "destinationAddress": "Testname test$test street 123456$68766 HOCKENHEIM$GERMANY",
    "references": [
        {
            "referenceType_code": "TRACKING",
            "action_code": "ADD",
            "altKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY:0080045552"
            "validFrom": "2020-08-03T05:50:10Z",
            "validTo": "9999-12-30T23:00:00Z"
        }
    ],
    "itemNo": "000020",
    "billofLading": "",
    "orderQuantity": 1.000,
    "executionStatus_code": "NOT_STARTED",
    "shippingPoint": "0001",
    "deliveryAltKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY:0080045552"
    "podStatus_code": "X",
    "weightUoM": "KG",
    "goodsIssueStatus_code": "A",
    "warehouseDescription": "",
    "quantityUoM": "EA",
    "pickingStatus_code": "C",
    "actualBusinessTimeZone": "CET",
    "upc": "5007509",
    "priority": 0,
    "volume": 1.000,
    "netWeight": 4.000,
    "incotermsVersion": "0001",
    "salesOrderItem_id": "42fdd0f4-a566-5b66-8e0c-d3d27dbb0272",
    "destinationTelephone": "13912341234",
    "materialNo": "HST_F1",
    "departureAddress": "Company$Dietmar-Hopp-Allee 16$69190 WALLDORF$GERMANY",
    "delivery_id": "56dcfddc-555c-5209-b7e8-fc863aaa5cf2",
    "destination": "0000010105",
    "destinationEmail": "test@163.com",
    "altKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY_IT:0080045552000020
    "modelNamespace": "com.lbngttapps.gtt.app.gttft1",
    "volumeUoM": "FT3",
    "actualTechnicalTimestamp": "2020-08-03T05:50:16.538Z",
    "dangerousGoods": "false",
    "deliveryNo": "0080045552",
    "plannedEvents": [
        {
            "plannedBusinessTimestamp": "2020-08-04T22:00:00Z",
            "plannedBusinessTimeZone": "CET",
            "plannedTechnicalTimestamp": "2020-08-04T22:00:00Z",
            "eventType": "com.lbngttapps.gtt.app.gttft1.Delivery.GoodsIssued",
            "locationAltKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:Location:ShippingPoin
```

```
            }
        ],
        "incoterms_code": "CIF",
        "materialDescription": "Laptop",
        "warehouseNo": "",
        "actualBusinessTimestamp": "2020-08-03T05:50:10Z",
        "grossWeight": 5.000,
        "salesOrderItemAltKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:SALES_ORDER_ITEM:000007
        "incotermsLocation1": "",
        "packingStatus_code": "X",
        "revisedPlannedDate": "2020-08-06",
        "messageSourceType": "TM",
        "myattachments": [
            {
                "fileName": "POD.png"
            },
            {
                "fileName": "PrivatePic.png",
                "privateAttachmentURL": "<URL to your private file>"
            }
        ]
    }
    attachment[]:file list
```

ⓘ Note

When sending “POST” requests, set the request body to “form-data” and add two key-value pairs:

Key	Type	Value
event	Text	Copy the payload of “event”
attachments[]	File	Select the files with the same file name you defined in “myattachments”. They can't be private URLs.

- In this example, “`myattachments`” is an event field with the type “Attachments” you defined in your model. You can have user-defined names.
- You must **upload** at least one file. You can also add private URLs to attachments with `privateAttachmentURL`. However, sending a request with private attachment URLs only is not supported. You can use the JSON Format for such case.

JSON Format

Request Header

Set the `LBN-GTT-Input-Data-Attachments-Flag` to true.

Request Body

Here is the sample request body:

‹≡› Sample Code

```
{
    "altKey": "xri://sap.com/id:LBN#10002424:Q8JCLNT774:lcTrakcingID:1691739430",
    "actualBusinessTimestamp": "2023-07-15T16:21:00.213+01:00",
    "actualBusinessTimeZone": "CET",
    "myattachments": [
        {
            "fileName": "POD.png",
            "fileContentBase64": <Base64Code>
        },
        {
            "fileName": "PrivatePic.png",
            "privateAttachmentURL": "<URL to your private file>"
        }
    ]
}
```

ℹ Note

- Base64 is a binary-to-text encoding scheme that is commonly used to represent binary data (such as images or files) in a format that can be transmitted over text-based protocols, such as email or HTTP.
- The file specified within "`privateAttachmentURL`" is referenced by a URL that points to a file located within your internal network.
- In this example, "`myattachments`" is an event field with the type "Attachments" you defined in your model. You can have user-defined names. You can attach files by uploading them in base64 format, providing private attachment URLs, or using a combination of both methods.

Add an additional attachment to an existing tracked process with attachments

You can add an additional attachment to an existing tracked process that has attachments in two formats: form-data format or JSON format.

Form-data Format

Request Header

`Content-Type: multipart/form-data`

Request Body

In this case, the request body is in form-data format. Keep the `fileId` of the attachment that you want to add. Here is the sample request body:

⁖≡› Sample Code

```
event:
{
    "altKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY_IT:0080045552000020
    "actualBusinessTimestamp": "2020-08-03T05:50:10Z",
```

```
    "myattachments": [
        {
            "fileId": "AOblXaaJOIKKGrbqloy888Qgyr9YkOl6tHWpd3bhV11"
        },
        {
            "fileName": "UnloadingInstruction.png"
        },
        {
            "fileName": "PrivatePic.png",
            "privateAttachmentURL": "<URL to your private file>"
        }
    ]
}
attachments[]: file list
```

ℹ Note

When sending “POST” requests, set the request body to “form-data” and add two key-value pairs:

Key	Type	Value
event	Text	Copy the payload of “event”
attachments[]	File	Select the files with the same file name you defined in “myattachments”. The files can't be private URLs.

- In this example, “`myattachments`” is an event field with the type “Attachments” you defined in your model. You can have user-defined names.
- You must upload at least one file. You can also add private URLs to attachments with privateAttachmentURL. However, submitting a request with private attachment URLs only is not supported. You can use the JSON format for such case.

JSON Format

Request Header

Set the `LBN-GTT-Input-Data-Attachments-Flag` to true.

Request Body

Here is the sample request body:

‹≡› Sample Code

```
event:
{
    "altKey": "xri://sap.com/id:LBN#10002424:Q8JCLNT774:lcTrakcingID:1691739430,
    "actualBusinessTimestamp":  "2023-07-15T16:21:00.213+01:00",
    "actualBusinessTimeZone": "CET",
    "myattachments": [
        {
```

```
            "fileId": "AOblXaaJOIKKGrbqloy888Qgyr9YkOl6tHWpd3bhV11"
        },
        {
            "fileName": "UnloadingInstruction.png"
            "fileContentBase64": <Base64Code>
        },
        {

            "fileName": "PrivatePic.png",
            "privateAttachmentURL": "<URL to your private file>"
        }
    ]
}
```

ℹ Note

- Base64 is a binary-to-text encoding scheme that is commonly used to represent binary data (such as images or files) in a format that can be transmitted over text-based protocols, such as email or HTTP
- The file specified within “privateAttachmentURL” is referenced by a URL that points to a file located within your internal network.
- In this example, “myattachments” is an event field with the type “Attachments” you defined in your model. You can have user-defined names.
- You can attach files by uploading them in Base64 format, providing private attachment URLs, or using a combination of both methods.

Update a tracked process to remove attachments

You can remove an attachment in two ways.

Option 1

Keep the `fileId` of the attachment that you want to have, and remove the others.

Request Body

Sample Code

```
{
    "altKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY_IT:0080045552000020
    "actualBusinessTimestamp": "2020-08-03T05:50:10Z",
    "myattachments": [
        {
            "fileId": "AOblXaaJOIKKGrbqloy888Qgyr9YkOl6tHWpd3bhV11",
            "fileName": "POD.png"
        }
    ]
}
```

Results

- Before the request, there are three files: “POD.png” and “UnloadingInstruction.png” and "PrivatePic.png".
- After the request, the file “UnloadingInstruction.png” and “PrivatePic.png” are deleted.

ℹ Note

For private URLs, you can get the `fileId` in two ways:

- In the Manage Message Logs apps, you can locate it in the **Payload** tab of the request's **Event** process flow.
- In the Read Service API, expand the attachment field you have defined and you can see the `fileId`.

Option 2

Use the property “action” to delete an attachment. Only the “DELETE” action is supported.

Request Header

Set the `LBN-GTT-Input-Data-Attachments-Flag` to true.

Request Body

Sample Code

```
{
    "altKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY_IT:0080045552000020
    "actualBusinessTimestamp": "2020-08-03T05:50:10Z",
    "myattachments": [
        {
            "fileName": "UnloadingInstruction.png",
            "action": "DELETE"
        }
    ]
}
```

Results

- Before the request, there are three files: “POD.png” and “UnloadingInstruction.png” and "PrivatePic.png".
- After the request, the file “UnloadingInstruction.png” is deleted.

Request Body Properties

Property	Type	Mandatory	Description
references	*Array of tracking Ids*	*no*	*The additional tracking Ids of the tracked process.*
altKey	*String*	*yes*	*The alternative key.* *Assuming URN style keys we have as example first our EM-Style keys:* *xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY:*

Property	Type	Mandatory	Description
			as string serialization with structural components: • *xri://sap.com/id: This prefix is used to identify documents/processe system* • *LBN#10010001016: an id for solution owner or solution participant* • *QM7CLNT910: Business system name* • *OUTBOUND_DELIVERY: the document's type. This is TrackingIdType* • *0080045552: The 'local' number range key*
validFrom	*Timestamp*	*no*	*The start date and time as local timestamp from which a tracking id is valid particular tracked process. Supported timestamp format: yyyy-MM-ddTHH:*
validTo	*Timestamp*	*no*	*The end date and time as local timestamp until when a tracking id is valid fo tracked process. Supported timestamp format: yyyy-MM-ddTHH:mm:ssZ*
referenceType_code	*String*	*yes*	*Fixed: TRACKING*
action	*String*	*yes*	*It can be only set to ADD/DELETE:* • *· ADD - creates/updates tracking id* • *· DELETE – deletes tracking id*
plannedEvents	*Array of planned event*	*no*	*Information about planned events*
eventType	*String*	*yes*	*The type that identifies the type of the event*
locationAltKey	*String*	*no*	*The alternative key of the location where the event is expected to happen*
eventMatchKey	*String*	*no*	*Identifier for planned event*
longitude	*Decimal*	*no*	*The longitude where planned event will happen*
latitude	*Decimal*	*no*	*The latitude where planned event will happen*
messageSourceType	*String*	*no*	*Use the field to identify which source system the event comes from*
payloadSequence	*Integer*	*no*	*The event sequence*
isFinalPlannedEvent	*Boolean*	*no*	*This need to be set true for final event*
plannedTechnicalTimestamp	*Timestamp*	*no*	*The timestamp with timezone when the event is expected to be reported to Business Network Global Track and Trace. Supported timestamp format: yy ddTHH:mm:ssZ*
plannedBusinessTimestamp	*Timestamp*	*no*	*The timestamp with timezone when the event is expected to actually happe timestamp format: yyyy-MM-ddTHH:mm:ssZ*
plannedBusinessTimeZone	*String*	*no*	*The time zone which is mapping to java time zone*
altKey	*String*	*yes*	*Same to references.altKey*
locationAltKey	*String*	*no*	*The alternative key of the location*
eventMatchKey	*String*	*no*	*Identifier for planned event*